EL NIÑO ACTION BRIEFING · 13 OCTOBER 2026

# Munjoy Hill, Portland, Maine

*In Maine, the El Niño winters people remember brought clustered storms: the 1998 ice storm, and in 2023–24 an outage storm followed by Portland's highest tide on record. The deaths that could be prevented came from carbon monoxide in the outages, not from the water. On Munjoy Hill, the person most at risk is an older renter alone on the third floor of a triple-decker. So the first job is to get a working CO alarm into every home that lacks one, give every household a warning in a language it reads, and organise a knock on that third-floor door before the first storm.*

| IMPACT REGION | EXCESS HEAT DEATHS | WINDOW | DOMINANT HAZARD |
|---|---|---|---|
| Cumberland County, ME (inferred, ~300,000 people) | None published for the US, Maine, or the county | n/a here; global 6-month figure 239,000 ± 11,000 | Clustered winter storms: multi-day outages, CO and cold homes; coastal surge on the waterfront |

**10/10 recommendations checked.** 8 confirmed after correction, 0 confirmed as written, 1 unverified (backup generators for food pantries), and 1 contraindicated (a parametric flood-insurance pilot this winter). The five funding placements were checked separately: 3 partially confirmed and 2 unverified. **The funding target fell 55%**, from $495k to $225k. That is more than the one-third flag, for three reasons: the warming shelter is already funded for 2026–27, the per-alarm cost had no source, and the fuel line was resized to what the existing fund moves. Exposure: Climate Impact Lab, 451,000 at Risk,

EXECUTIVE SUMMARY — AT A GLANCE




43.67°N 70.25°W, United States of America · Natural Earth

- **The risk.** Clustered storms are the pattern, as in 1998 and 2023–24, and the preventable deaths come from carbon monoxide in the outages, not from the water.
- **First action.** Portland Fire, the Red Cross, and immigrant-led groups put a working CO alarm in every home that lacks one, with a picture warning in each household's language, by 1 December.*
- **The money.** About $55k floor, $225k target, $515k ceiling; the floor starts with $35k for alarms in 500 homes and $15k for In Her Presence outreach.
- **Do not.** Do not make "buy a generator" the headline outage advice: all 18 CO poisonings after the December 2023 storm came from generators in garages or basements.*


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* Suggested formulation: check with a health worker before sharing.

September 2026, pre-peer-review. It publishes no figure for the United States, Maine, or Cumberland County, and heat is not this place's winter hazard.

**Scope: built for Munjoy Hill, from a request naming one street.** A single street is finer than any source publishes data for, so this briefing covers the neighbourhood around it: Munjoy Hill and the East End (ZIP 04101), plus Franklin Towers, Kennedy Park, and the Commercial Street waterfront next to them. Statistics for the city or the county are labelled as such.

**Read this first: El Niño is a weak guide here, and the plan should outlive it.** NOAA gives a strong El Niño through the winter, but forecasters disagree about which way its New England signal points. What the record shows is clustering. Portland's two highest tides in 112 years both came in El Niño winters: 14.17 ft above mean lower low water in February 1978 (a weak event) and 14.57 ft on 13 January 2024 (a strong one). The strong 1997–98 El Niño brought the January 1998 ice storm, which produced Maine's worst recorded carbon monoxide outbreak. The strong 2023–24 El Niño brought the 18 December storm, with 18 CO poisonings, all from generators, followed by the January surge. But the record October 2017 windstorm came in a neutral year. So prepare now for outages and CO, and keep every measure in place after March. The Skeptic's warning stands: a plan labelled “El Niño” is a plan that stands down when the label expires.

## WHAT THIS PLACE SHOULD DO

RANKED

Actions are ranked by expected lives saved per unit of effort before the peak. The backing lenses are shown on each action, and the RACI chart below assigns each one. An action several lenses reached independently is the stronger bet, but only where its evidence came from somewhere

* Suggested formulation: check with a health worker before sharing.

like here.

## 1. Put a working CO alarm in every home that lacks one, and give every household a picture warning in its own language: generator outside only; no grill, charcoal, or gas oven for heat*

PARTIALLY CONFIRMED | EMERGENCY MGR · PUBLIC HEALTH · FRONTLINE · SKEPTIC · HISTORIAN · PHILANTHROPIST · FOOD | INVENTORY: NONE FOUND

| OWNER | BY WHEN | COST | PREVENTS |
|---|---|---|---|
| Portland Fire, with the Red Cross of Northern New England; In Her Presence and other immigrant-led groups carry the messages | 1 Dec, before the first outage | ~$70k for alarms, ~$30k for outreach (see funding) | CO poisoning in multi-day outages, the main preventable killer in 1998 and 2023 |

In the January 1998 ice storm, Maine recorded 100 CO poisonings from 42 incidents. Generators caused 30 of the incidents, and 1 person died. Running the generator in a basement or attached structure was the strongest risk factor (Daley et al., *J Emerg Med* 2000). In a CDC survey of households without power, only 3 of 36 (8%) had a working CO alarm. The 18 December 2023 storm repeated the pattern: Maine CDC logged 18 poisonings from 8 incidents, all from generators in garages or basements, and none fatal. Yoon et al. (*JAMA* 1998) judged that alarms could have prevented about half of 136 New Mexico CO deaths.

**Where the gap is.** Maine's alarm law (25 MRSA §2468) already covers every rented unit, single-family rentals included. So for Munjoy Hill's renters the job is to check that the alarms exist and work. The installation gap is owner-occupied homes acquired on or before 31 October 2009. Most households here rent triple-decker units and have nowhere safe to run a generator, so the message leads with charcoal, grills, and gas ovens, and generators come second. In Seattle, 85% of indoor-charcoal CO cases with race recorded were non-White, and 55% of households had a member whose main language was not English (Gulati et al., *AJPH* 2009). Maine CDC publishes CO material in English only. US CDC publishes Arabic and French versions, and Vermont publishes a Somali one, so translations exist and can be reused. Send them as WhatsApp voice notes through community groups as well as on paper.

* Suggested formulation: check with a health worker before sharing.

Transferability: the evidence is from Maine, for the same hazard. The Seattle language finding comes from another city and is a risk signal, not a documented Portland case. Corrections made in verification: the lens's “73% minority” Seattle figure was wrong, since 73% was the county's White baseline. No Red Cross Sound the Alarm event or Portland Fire install programme is currently scheduled for Portland, so this needs one to be organised. 14 MRSA §174 allows fire departments to install alarms. Sealed 10-year units retail at about $40–60.

## 2. Knock on the third-floor door: a consent-based paper roster of older, isolated, and power-dependent residents, each paired with a neighbour, and visited within 12 hours of losing power

PARTIALLY CONFIRMED | EMERGENCY MGR · PUBLIC HEALTH · FRONTLINE · SKEPTIC · HISTORIAN · PHILANTHROPIST | INVENTORY: NONE FOUND

| OWNER | BY WHEN | COST | PREVENTS |
|---|---|---|---|
| **Munjoy Hill Neighborhood Organization (MHNO), with volunteers; Portland Housing Authority (PHA) floor captains at Franklin Towers** | **Roster by 30 Nov** | **$5k pilot, through a fiscal sponsor (see funding)** | **Older and device-dependent residents left alone in a cold, dark flat for days** |

No vulnerable-persons register was found at the city, at Cumberland County EMA, or at MEMA. Munjoy Hill is mostly triple-deckers, and gentrification and short-term rentals have left long-time renters ageing in place next to units that sit empty all winter. The Frontline lens put it plainly: the person who dies in a Hill outage is the 80-year-old on the third floor whose neighbours are now Airbnbs. MHNO is active and has a 2025–26 board, so it can hold the roster. At the foot of the Hill, Franklin Towers is a 16-storey PHA building for older and disabled tenants. From 26 August to 22 September 2022, its floors 7–16 ran on reduced power, and PHA handed out extension cords for medical devices. Ask PHA to name a resident floor captain on each floor who knocks on doors during any outage longer than 12 hours. Device users should enrol in CMP's Lifelight programme. It gives advance notice of outages, but not priority restoration, so do not tell anyone it does. Use HHS emPOWER's count for ZIP 04101 to size the roster.*

* Suggested formulation: check with a health worker before sharing.

Transferability: the mortality evidence comes from a summer blackout in a city. In New York in 2003, accidental deaths rose 122% (95% CI 28–287%) and non-accidental deaths rose 25% (Anderson & Bell, *Epidemiology* 2012). That is weak evidence for a Maine winter. The lens's figure that 37.4% of Portland residents over 65 live alone could not be re-checked, because the source PDF is offline. Kennedy Park is low-rise family housing, so it needs block contacts rather than floor captains.

## 3. Make sure the downtown warming shelter actually opens every night: sign the Seeds of Hope contract and name a staffing backup in writing

PARTIALLY CONFIRMED | EMERGENCY MGR · PUBLIC HEALTH · SKEPTIC · PHILANTHROPIST | INVENTORY: NONE FOUND

| OWNER | BY WHEN | COST | PREVENTS |
|---|---|---|---|
| City of Portland Health & Human Services and City Manager; Seeds of Hope as operator | Contract before 15 Nov (MaineHousing funding starts then) | Funded: $360,720 from MaineHousing | A repeat of the nights in Dec 2025 and Feb 2026 when the shelter did not open |

Portland's shelter failed to open twice last winter for lack of staff: on a night below 15°F in December 2025, and during the February 2026 blizzard. The shelter then ran on city staff working overtime. For 2026–27, MaineHousing has awarded Seeds of Hope, a Biddeford nonprofit, $360,720 to run a downtown shelter every night “regardless of temperature or snowfall.” That removes the trigger problem, but no signed city contract was found, and the mayor has raised concerns about capacity. The Emergency manager's point stands: a shelter is only open if someone is contractually obliged to unlock the door. Name the backup operator and the backup staff list in the contract. Preble Street counted 52 deaths among unhoused people in Portland in 2025 and 45 in 2023, so the base rate is about one a week.

* Suggested formulation: check with a health worker before sharing.

Transferability: native. No outcome study shows that warming shelters reduce deaths; the evidence is descriptive. Corrections made in verification: the December 2024 shelter was at First Parish on Congress St, not on Riverside. The lens's Riverside shuttle was dropped because the shelter is moving downtown.

## 4. Designate East End Community School as the Hill's warm room during outages: check its backup power and publish when it opens

PARTIALLY CONFIRMED | HISTORIAN · SKEPTIC · EMERGENCY MGR | INVENTORY: NONE FOUND

| OWNER | BY WHEN | COST | PREVENTS |
|---|---|---|---|
| City of Portland emergency management, with Portland Public Schools | Site and threshold published by 1 Dec | Staff time; generator cost unknown; *unfunded* | Residents without a car having nowhere warm to walk to during an outage |

There is a precedent. On 24–25 March 2024, with about 15,400 CMP customers out, the city opened an overnight warming centre at East End Community School (195 North St), and METRO waived fares to reach it. The Hill's library branch closed in 2010, so the school is the only public room on the Hill. The action: confirm whether the school has backup power, and publish the opening rule. The Skeptic proposed opening at 32°F or during any outage longer than 12 hours. That threshold no longer matters for the shelter, which will open every night, but it is the right one for this warm room. A Toronto study found most cold injuries happen in moderate cold, not extreme cold.

* Suggested formulation: check with a health worker before sharing.

Transferability: native precedent. Not verified: whether the school has a generator. Without one, the room is warm only while the school's own power holds.

## 5. Get households to heat: HEAP is open now; ECIP from 1 November for anyone running out of fuel; and the CMP arrearage and Lifeline programmes for electricity debt

PARTIALLY CONFIRMED | ECONOMIST · FOOD · PHILANTHROPIST · PUBLIC HEALTH | INVENTORY: NONE FOUND

| OWNER | BY WHEN | COST | PREVENTS |
|---|---|---|---|
| The Opportunity Alliance (Cumberland County's community action agency); Root Cellar, MHNO, and In Her Presence for sign-up help; CMP and the Maine PUC | HEAP now; ECIP from 1 Nov | Benefits already funded; outreach time | Rationed heat, and food money spent on oil |

Heating oil averaged $5.97 a gallon statewide on 5 October 2026 (Maine Department of Energy Resources weekly survey; range $5.45–6.81). HEAP for 2026–27 opened on 3 August, and The Opportunity Alliance handles intake for Cumberland County. ECIP, the crisis benefit for households nearly out of fuel, runs from 1 November 2026 to 30 April 2027, pays a one-time maximum of $500 (down from $800 in 2024–25), and requires a HEAP application first. For electricity, CMP's Arrearage Management Program forgives up to $500 of a qualifying past-due balance each month, to a cap of $6,000 a year, and its Electricity Lifeline Program gives up to $1,200 a year according to CMP. Both work through HEAP eligibility. The Food lens's point is that when oil reaches $6, the food bank becomes the heating-oil subsidy. Every household signed up for HEAP takes that pressure off its food budget.

* Suggested formulation: check with a health worker before sharing.

Transferability: native. Corrections made in verification: the winter disconnection rule is 35-A M.R.S. §717, not §718. It runs from 15 November to 15 April. It is a permission gate (a utility needs PUC approval to disconnect), not a ban, and it covers electricity and gas, not oil. A lens's claim of an early $36M LIHEAP release could not be found in the Governor's release, so it is dropped. That release does confirm that the President's budget eliminates LIHEAP, which puts next winter at risk, not this one.

## 6. Hold private fuel money in reserve, released when ECIP runs out or during a multi-day outage

PARTIALLY CONFIRMED | ECONOMIST · PHILANTHROPIST | INVENTORY: NONE FOUND

| OWNER | BY WHEN | COST | PREVENTS |
|---|---|---|---|
| **United Way of Southern Maine (Keep ME Warm), paid out through The Opportunity Alliance** | **Release rule agreed by 1 Dec** | **$120k, half held as a reserve (see funding)** | **Households out of oil after the federal benefit is spent** |

Keep ME Warm is the private fuel fund that United Ways and community action agencies have run since 2008, for households who have exhausted every other option. It raised $740,445 in 2023, and half its money already flows through community action agencies, so this is an existing channel. The Economist's finding is that nothing here is pre-arranged: the fund pays out case by case, with no trigger. Last winter the federal shutdown kept ECIP closed until 3 December. The action is a written rule agreed in advance: release the reserve if ECIP money is exhausted, or if outages in Cumberland County last more than 48 hours. Private money cannot be added to ECIP itself, so it has to go through Keep ME Warm.

* Suggested formulation: check with a health worker before sharing.

Transferability: native. Correction made in verification: the lens's trigger (“if HEAP is not open by 15 November”) could never fire, because HEAP opened on 3 August. It has been recast.

## 7. Write down the waterfront trigger: close Commercial St, Marginal Way, and Somerset St when an NWS coastal flood warning coincides with a high tide, and brief wharf workers before the January spring tides

PARTIALLY CONFIRMED | EMERGENCY MGR · HISTORIAN · FRONTLINE · FOOD

INVENTORY: CHANGE IN FORECAST QUALITY (SHRADER ET AL. 2023), INDIRECT

| OWNER | BY WHEN | COST | PREVENTS |
|---|---|---|---|
| Portland Public Works, with NWS Gray | Written trigger by 1 Dec; wharf briefing before the January spring tides | Staff time | People and vehicles caught in a repeat of January 2024 |

The Portland gauge (NOAA 8418150) reached 14.57 ft above mean lower low water at 12:06 on 13 January 2024, beating 1978's 14.17 ft. NWS still marks the reading as preliminary. Water flooded Commercial Street, Portland Pier, the Old Port, and Back Cove, and the storms were declared DR-4764 with Cumberland County included. The city already barricades these streets at high tide, but no written trigger was found. Use NWS Gray's own flood categories for the gauge: 12, 13, and 14 ft for minor, moderate, and major. Treat 13 ft as the line for closures and for warning wharf crews to move gear and vehicles. The $3.6M Commercial Street outfall and tidal backflow-valve project is under construction from July 2026 to May 2027, so this winter meets the old drains.

* Suggested formulation: check with a health worker before sharing.

Transferability: native. This protects property and responders more than residents' lives, because no deaths were found in the 2024 surge. It is cheap. Correction made in verification: a lens's 13.5 ft threshold is not an NWS category. The breach at Chandler's Wharf may date from the 10 January storm rather than the 13th.

## 8. Stage three-day no-cook food boxes in the East End for households that lose power and cannot cook or keep food cold

PARTIALLY CONFIRMED | FOOD | INVENTORY: NONE FOUND

| OWNER | BY WHEN | COST | PREVENTS |
|---|---|---|---|
| **Good Shepherd Food Bank, with The Root Cellar (94 Washington Ave) and Preble Street** | **1 Dec** | **~$25k for ~1,000 boxes at ~$25 (unit cost unverified); *unfunded*** | **Spoiled food, and cooking indoors on combustion appliances during outages** |

Around here, food runs short at the household end, not at the source. When the power goes, the fridge is the first thing to fail, not the trucks on I-95. The supply side is also thinner than it was. Since March 2025, Good Shepherd's partner pantries have received half or less of their previous allocations. About 600 agencies depend on it, more than 75% of them run entirely by volunteers. SNAP benefits were halted on 1 November 2025 and paid in full only on 10–14 November, after a court order. USDA ERS forecasts food-at-home prices up 2.4% in 2026: ordinary inflation, with no El Niño component. Boxes in kind make sense only for the no-power, no-cooking case. Otherwise cash or SNAP is better, because shops are close by.

* Suggested formulation: check with a health worker before sharing.

Transferability: native. Corrections made in verification: no pantry was found at East End Community School, so the boxes go through The Root Cellar, the East End's emergency pantry (phone 207-774-3197 to confirm it is active in 2026). The lens's figures were wrong in two places: Maine did not pay November 2025 SNAP on time, and the 2.8% price forecast was May's; September's is 2.4%. SNAP enrolment fell to about 156,000 Mainers by March 2026.

## 9. Check backup power at the main food pantries and meal kitchens before funding generators

UNVERIFIED FOOD INVENTORY: NONE FOUND

| OWNER | BY WHEN | COST | PREVENTS |
|---|---|---|---|
| **Preble Street, Wayside Food Programs, and The Root Cellar** | **Survey by 15 Nov** | **Survey: none. Generators: no sourced figure; *unfunded*** | **Losing refrigerated and frozen stock in an outage** |

The lens's $10–30k per site had no source. Preble Street's central kitchen in South Portland already freezes up to 50,000 meals, so find out what backup power exists before anyone pays for more.

Transferability: native. Ranked last because nothing about its cost or need could be verified.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

## RACI CHART

**R** = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

| Action | City of Portland (HHS, EM, City Manager) | Portland Fire | Public Works | Red Cross NNE | Portland Housing Authority | MHNO | In Her Presence & immigrant groups | Seeds of Hope | Portland Public Schools | The Opportunity Alliance | United Way of S. Maine | CMP & Maine PUC | Good Shepherd, Root Cellar, Preble St, Wayside | NWS Gray |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. CO alarms and multilingual warnings | C | A/R | | R | R | C | R | | | I | | | I | |
| 2. Third-floor knock roster | C | C | | | R | A/R | C | | | | | I | I | |
| 3. Nightly shelter contract and backup | A | I | | | | | I | R | | | | | C | I |
| 4. East End Community School warm room | A/R | C | | C | | I | I | | R | | | I | | I |
| 5. HEAP, ECIP, AMP and Lifeline sign-up | I | | | | C | R | R | | | A/R | I | R | R | |
| 6. Keep ME Warm reserve and release rule | I | | | | | | | | | R | A/R | | I | |
| 7. Written waterfront closure trigger | C | C | A/R | | | I | | | | | | | | C |
| 8. No-cook boxes in the East End | I | | | | C | C | C | | | | I | | A/R | |
| 9. Pantry backup-power check | I | | | | | | | | | | | | A/R | |

What the chart shows. **The city is accountable for the two shelter rows** and is consulted or informed on almost everything else. Its job this winter is contracts and published rules, not new money. **Both of the top two rows sit with organisations that are not city agencies.** The knock roster sits with a volunteer neighbourhood association, which is deliberate, because people answer the door to a neighbour. **Rows 3, 4, and 7 need a decision, not money:** a signed contract, a published threshold, and a written trigger.

WHERE MONEY DOES THE MOST

## FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs. The warming shelter and the waterfront are excluded, because MaineHousing and MaineDOT already fund them.

FLOOR

~$55k

TARGET

~$225k

CEILING

~$515k

LARGEST UNCERTAINTY

How many East End households run out of fuel once ECIP's $500 is spent,

* Suggested formulation: check with a health worker before sharing.

| | |
|---|---|
| | **and whether the city's own shelter nights are unfunded** |

**Floor:** $35k for alarms (500 homes) + $15k for In Her Presence outreach + $5k for the MHNO roster = **$55k**. **Target:** $70k for alarms (1,000 homes × ~$70, an estimate: a $40–60 retail unit plus about $25 to install) + $120k for fuel (200 fills × $600, where 100 gal × $5.97 ≈ $600; half held as a reserve) + $30k for outreach + $5k for the roster = **$225k**. **Ceiling:** the target + $35k (500 more alarm homes) + $60k (100 more fills) + $25k (no-cook boxes, action 8) + $170k (a contingent city match for extra shelter nights, only if the 2026–27 nights prove unfunded; the $170k is 20 nights × $8.5k, which is a division of last winter's $400k ÷ 47 nights, not a city figure) = **~$515k**. **The target fell 55%** from the lens's $495k. The $170k city match moved to the ceiling because Seeds of Hope already holds $360,720. The alarm count was cut because the $70 unit cost had no published source. The fuel line was resized to what Keep ME Warm's existing channel moves.

| RECIPIENT | USE | AMOUNT | MECHANISM | BASIS | VERDICT |
|---|---|---|---|---|---|
| Red Cross of Northern New England with Portland Fire | Action 1: ~1,000 sealed combination smoke/CO alarms for long-held owner-occupied homes and failed rental units | $70k (was $105k) | Pre-positioned stock plus a grant to an existing free-install programme | A modelled home-visit smoke-alarm programme costs $28,252 per death averted (*Injury Prevention* 20(4):251, a US model, not an observed figure). The Red Cross installed 168 alarms in one day at a 2023 Knox County event. The unit cost is an estimate. | PARTIALLY CONFIRMED |
| The Opportunity Alliance via Keep ME Warm (United Way of Southern Maine) | Actions 5–6: emergency fuel once ECIP runs out; half held on a written release rule | $120k (was $180k) | Top-up to an existing private fund, held as a reserve that a trigger releases | Cheaper heating reduces winter deaths, mostly cardiovascular and respiratory, and more so in poor counties (Chirakijja, Jayachandran & Ong, NBER w25681). That finding is identified from gas and electricity prices, so applying it to oil heat is an analogy. Keep ME Warm raised $740,445 in 2023. | PARTIALLY CONFIRMED |

* Suggested formulation: check with a health worker before sharing.

| RECIPIENT | USE | AMOUNT | MECHANISM | BASIS | VERDICT |
|---|---|---|---|---|---|
| In Her Presence (4 Milk St) and other immigrant-led groups | Action 1 messages and action 5 sign-up, in Somali, Arabic, French, Portuguese, and Lingala | $30k, contingent | Direct grant | Active, and partnered with Preble Street in June 2026, with outreach to 843 women. $50 per household is an estimate, not a published cost. No Somali association was named, so that part was not checked. | PARTIALLY CONFIRMED |
| Munjoy Hill Neighborhood Organization, through a fiscal sponsor | Action 2: roster volunteers' costs and printing | $5k pilot (was $10k) | Small grant | Founded 1979 and active, but no record of handling grants was found. No cost-effectiveness study exists for this. | UNVERIFIED |
| City of Portland | Action 3: extra shelter nights | $0 placed; $170k in the ceiling only | Matching grant, contingent | Seeds of Hope holds $360,720 for nightly operation. The city is not named in the 2026–27 MaineHousing awards. Get its budget before matching anything. | UNVERIFIED |

**The funder's one thing:** Maine records only a handful of CO deaths a year statewide, so this envelope mostly buys harm averted, not a countable number of lives. Do not spend the fuel line in October. Money released on a written trigger reaches the household that ran out in January; money given as a photograph-driven appeal arrives after the cold snap.

* Suggested formulation: check with a health worker before sharing.

# Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for this hazard and would cost lives, or waste the window, here.

## Make “buy a generator” the headline outage advice *

AVOID (SKEPTIC; SUPPORTED BY DALEY 2000 AND MAINE CDC 2023)

All 18 CO poisonings after the December 2023 storm came from generators in garages or basements. In 1998, generators caused 30 of 42 CO incidents, and basement placement was the strongest risk factor. Most Munjoy Hill households rent a unit in a triple-decker and have no safe outdoor place to run one. Generator advice without the placement rule and an alarm is the advice that produced those poisonings.

**Instead:** lead with a working CO alarm and a warm room to walk to (actions 1 and 4). Every generator message carries “outdoors, away from windows; never in a basement, garage, or porch.”

## Launch a parametric flood-insurance pilot for this winter

CONTRAINDICATED FOR THIS WINTER

The Economist proposed a pilot paying $1,000 per household when the Portland gauge reaches 14.0 ft, funded by the Maine Community Foundation. No such product exists in Maine. A foundation is not a licensed insurer, and the Bureau of Insurance filing time puts the January–March peak out of reach. The trigger has been crossed twice since 1912, so it would almost never pay. It would spend the window on paperwork.

* Suggested formulation: check with a health worker before sharing.

**Instead:** if a funder wants money tied to the water, pre-commit cash transfers through Keep ME Warm or a community fund at NWS's 13 ft moderate flood level. Revisit cover for 2027–28. Only about 1% of Maine households hold flood insurance (FEMA, 7,452 policies in March 2024).

### Top up ECIP with private money

NOT POSSIBLE AS PROPOSED

ECIP is federal LIHEAP money run through the community action agencies, and donor money cannot be added to it. The same mistake was made, and caught, in the Rockport briefing.

**Instead:** route private fuel money through Keep ME Warm to The Opportunity Alliance (action 6).

### Count “hypothermia deaths”, and stand the plan down in March

AVOID (PUBLIC HEALTH; SKEPTIC)

In Portland, winter deaths will be recorded as overdoses, cardiac events, falls, and CO poisoning, and only rarely as hypothermia. A hypothermia count will always look reassuring. Maine's clustered storms run late: the March 2024 outage opened the East End warming room.

**Instead:** judge the winter by all-cause deaths among unhoused and isolated people from December to April, against Preble Street's baseline of about 50 a year. Keep every measure in place until April.

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

| YEAR | EVENT | TOLL | WHAT CHANGED AFTER |
|---|---|---|---|
| 4 Jul 1866 | Great Fire: about 1,500–1,800 buildings burned, much of the peninsula including the East End | 2–4 deaths (sources differ); 10,000+ left homeless | Rebuilt in brick; the Hill's wooden triple-deckers came later |

* Suggested formulation: check with a health worker before sharing.

| YEAR | EVENT | TOLL | WHAT CHANGED AFTER |
|---|---|---|---|
| Feb 1978 | Blizzard during a weak El Niño (DJF ONI +0.7); Portland tide record 14.17 ft MLLW | Not found for Portland | The record stood for 46 years |
| Jan 1998 | Ice storm during a strong El Niño (+2.2); up to 80% of Maine without power, some homes for weeks | 100 CO poisonings, 1 CO death (Daley); 3–8 total deaths depending on source | Statewide CO messaging; alarm law came in 2009 |
| Oct 2017 | Windstorm: Maine's largest outage at the time; ENSO-neutral | Not found | Shows that not every disaster here is an El Niño one |
| Aug–Sep 2022 | Franklin Towers electrical failure: floors 7–16 on reduced power for nearly four weeks | None reported | PHA handed out extension cords; no floor-captain system found |
| 18 Dec 2023 | Windstorm during a strong El Niño (+1.8): ~400,000+ Maine outages, CMP's strongest wind on record | A man killed by a falling tree in Windham (Cumberland Co.); 18 CO poisonings, none fatal | DR-4754, Cumberland not designated |
| 10 & 13 Jan 2024 | Surge storms; Portland record 14.57 ft MLLW (preliminary); Commercial St, Portland Pier, and Back Cove flooded | No deaths found in Portland | DR-4764 (Cumberland included); $25M for working waterfronts, $2M each to Maine, Custom House, and Widgery Wharves; outfall project, built 2026–27 |
| 24–25 Mar 2024 | Outage, ~15,400 CMP customers | None found | Overnight warming centre at East End Community School, with fares waived |
| Dec 2025 & Feb 2026 | Warming shelter did not open: a night below 15°F, and the February blizzard | Not reported | Blizzard and ice-storm triggers added; nightly operator funded for 2026–27 |

* Suggested formulation: check with a health worker before sharing.

| YEAR | EVENT | TOLL | WHAT CHANGED AFTER |
|---|---|---|---|
| 2023 / 2025 | Deaths among unhoused people in Portland (Preble Street vigil counts; scope may vary by year) | 45 / 52 | Seasonal shelter expansion |

**The recurrence nobody has noticed:** Portland's two highest tides in 112 years, 1978 and 2024, both came in El Niño winters. Each time the cause was a storm on top of a spring tide, not sea level alone, and each time the city answered with proposals, such as the $103M Back Cove surge barrier, rather than structures. Two cases do not prove El Niño causes these tides. But this is the winter to treat the January and February spring tides as the risk window.

## WHERE THE EXPERTS DISAGREE

UNRESOLVED

- **Prepare for this El Niño** (Historian, Emergency manager) versus **El Niño is a weak frame for Portland's winter** (Skeptic, Food). Both are right about different things. The 1998 and 2023–24 clusters support preparing now, and the Skeptic's point is about how long the plan lasts. That is why the contraindicated list includes standing down in March.
- **Target alarms at renters' triple-deckers** (Frontline, Emergency manager) versus **at long-held owner-occupied homes** (Philanthropist). The statute settles it. Rentals are already covered, so the job there is checking that alarms exist and work. New installs go to owner-occupied homes acquired before November 2009.
- **Open shelter at 15°F** (city practice until 2025–26) versus **at 32°F or during a 12-hour outage** (Skeptic). For the shelter this has been overtaken, because it now opens nightly. For the outage warm room the threshold still has to be chosen and published (action 4).
- **Food boxes in kind** (Food) versus **cash** (Philanthropist, Food verifier). In kind only for the no-power, no-cooking case. Cash or SNAP otherwise, because shops are close.

## WHAT NOBODY COVERED

BLIND SPOT

No lens found data on **heat pumps**, which stop in a power cut and which many Portland homes now rely on, so the outage plan may underestimate how many homes go cold. No lens produced **Portland-level CO, overdose, or cold-exposure counts**; the CO figures are statewide, and the overdose link is inferred. No lens examined **fire risk from space heaters in the Hill's wooden triple-deckers** with local data.

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The 2014 Noyes Street fire, which killed 6 people, was caused by smoking materials, not a heater. No current figure exists for **asylum seekers** in the shelter system; the last count found was from 2023.

HOW TO READ THIS METHOD

**The panel is author-built.** Eight simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are resolved at Climate Impact Lab impact-region scale, roughly 300,000 people. They are not neighbourhood figures. Cumberland County holds about 300,000 people, so it is a plausible match for an impact region, but it was inferred, not resolved on the Impact Map. The El Niño projection covers *heat only*. Outages, surge, and cold are carried here by the historical record, not by a published number.

HOW THIS BRIEFING WAS PRODUCED METHODOLOGY

- **Scope.** The request named a single street in Portland, ME. OpenStreetMap geocodes it to Munjoy Hill, ZIP 04101, Cumberland County. A single street is too fine for any data source, so the briefing covers Munjoy Hill and the East End, plus the neighbouring Franklin Towers, Kennedy Park (East Bayside), and the Commercial Street waterfront. Climate Impact Lab impact region, as this report records it: Cumberland County, ME, *inferred*. Where administrative data is used, the unit is named: the State of Maine (CO cases, outages, SNAP, oil prices, NFIP), Cumberland County (FEMA designations, HEAP intake), the City of Portland (shelter, unhoused deaths, the over-65 living-alone share), and ZIP 04101 (emPOWER). None is Munjoy Hill alone.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Where heat is not the dominant hazard, the historical record carries the hazard instead of a borrowed number.
- **Expert panel.** Eight simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, philanthropist, and food supply chain analyst) each researched this place independently and in parallel, from web sources, to a fixed brief. The Rockport, Maine briefing (8 October 2026) served as an evidence pack. Its claims were re-checked, not inherited: this run corrected its winter-disconnection statute (§717, not §718) and its framing of the oil price.

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- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias before it counted. The seven-lens convergence on CO alarms rests on a Maine outbreak study (1998) and a Maine CDC advisory (2023), so it held. The Philanthropist's CO cost figure did not: its source paper prices CO-only alarms. Contradictions were kept. Actions were ranked by expected lives saved per unit of effort, each was given a payer or marked unfunded, and the funding placements were reconciled against them.
- **Verification.** Six separate verifier agents checked 10 recommendations and the five funding placements for evidence, transferability, and feasibility. The financial actions had their own verifier, which checked availability, payer, trigger, and existing implementation. So did the food actions, checking trigger publication frequency, forecast timing, and channel capacity, and the envelope, checking real recipients, unit costs, and duplication, before it was re-summed.
- **Who does what.** The RACI chart is built from the owners named in the verified actions, with one accountable party per action. It is a proposal: no organisation named in it has been contacted or has agreed to a role.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Not consulted directly in this run. The CO verifier's fetch returned the empty client-side shell (“Number of Adaptations: 0”). The tags rely on the inventory's US filter (19 entries), read in a browser for the Gaslamp Quarter and Rockport briefings. No US entry covers CO alarms, check-in rosters, warming rooms, or fuel assistance. Only *Change in forecast quality* (Shrader et al. 2023) bears, indirectly, on the waterfront trigger.
- **Health wording.** Passages where a wording error could hurt someone (carbon monoxide, generators, CO alarms, and medical equipment in an outage) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total without the division being shown and labelled. The $8.5k shelter night ($400k ÷ 47) and the $600 fill (100 gal × $5.97) are shown and labelled. No local figure is invented; where data is thin, the report says so.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. The Maine CDC Portland City Health Profile returned 404, so the 37.4% over-65 living-alone figure is unverified. CMP's relief page timed out, so the AMP and Lifeline figures come from search extracts and CMP's March 2026 release. emPOWER's ZIP 04101 count was not re-checked. No Portland-specific storm death counts were found. The Root Cellar's 2026 status and East End Community School's backup power are unconfirmed.

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References · verified by Phase 5 unless marked. Daley et al. 2000, 1998 Maine CO outbreak (verified) · CDC MMWR 1998, CO alarm survey (verified) · Maine CDC advisory 2023PHADV045 (verified) · Yoon et al. 1998, JAMA (verified) · Gulati et al. 2009, Seattle charcoal CO (corrected) · 25 MRSA §2468 (verified) · Red Cross NNE CO alarm installs (verified) · Red Cross, 168 alarms in a day (verified) · Anderson & Bell 2012 (verified) · Franklin Towers outage, BDN 2022 (corrected) · Munjoy Hill Neighborhood Organization (verified) · CMP Lifelight (verified) · Portland shelter triggers, Mar 2026 (verified) · Shelter operator search, Jun 2026 (verified) · Shelter failed to open, Dec 2025 (verified) · MaineHousing 2026–27 shelter awards (verified) · Dec 2024 shelter (corrected) · Preble Street vigil 2025 (verified) · Unhoused deaths 2023 (verified) · City alert, EECS warming centre, Mar 2024 (verified) · Toronto cold-injury study (lens, verifier-cited) · MaineHousing HEAP (verified) · MaineHousing ECIP cap (verified) · Governor Mills on LIHEAP, 2 Oct 2026 (verified) · Maine DOER heating oil, 5 Oct 2026 (verified) · CMP AMP and Lifeline (partially verified) · Maine PUC winter disconnection, §717 (verified) · Keep ME Warm (verified) · Keep ME Warm 2023 total (verified) · ECIP delay, 2025 (verified) · Chirakijja, Jayachandran & Ong (verified) · In Her Presence (verified) · NWS Gray, Portland record tide (verified) · NOAA ONI v6 (verified) · FEMA DR-4764 (lens, not re-verified) · Commercial St outfall project (verified) · Portland high-tide road closures (verified) · Working waterfront aid, 2024 (lens, not re-verified) · FEMA Maine NFIP profile (verified) · Good Shepherd allocations, 2025 (verified) · November 2025 SNAP (verified) · USDA ERS Food Price Outlook (verified) · Maine state economist, SNAP cost shift (verified) · 2024 Portland flood (lens, not re-verified) · Great Fire of 1866 (lens, not re-verified) · Ice storm of 1998 (lens, not re-verified) · 2014 Noyes Street fire (lens, not re-verified)

Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory (US)

Generated 13 October 2026 for Munjoy Hill, Portland, Maine. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

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